
A Hybrid Framework for Hallucination Detection and Mitigation in Large Language Models

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1 Introduction

Large Language Models (LLMs) have achieved strong performance across many language understanding and reasoning tasks. However, they often generate incorrect or unsupported statements, a phenomenon known as *hallucination*. This issue limits the reliability of LLMs in applications where factual accuracy and trustworthy reasoning are essential.

Recent work has explored different directions to detect and mitigate hallucinations. One approach analyzes the internal computation of LLMs. Frasca et al. represent model attention patterns as attributed graphs and apply Graph Neural Networks (GNNs) to detect hallucination-related structures within attention flows. Another line of work focuses on uncertainty signals during generation. Zou et al. propose monitoring self-assessed confidence and token-level entropy to identify unstable reasoning and guide training through reinforcement learning.

Although both approaches provide valuable insights, they capture different aspects of the hallucination problem. Graph-based methods exploit structural relationships within model computations, while uncertainty-based approaches rely on local signals of reasoning instability. Treating these signals independently limits their ability to capture the full spectrum of hallucination behavior.

In this work, we propose a hybrid framework that combines structural and uncertainty-based signals for hallucination detection and mitigation. Our approach constructs attention graphs from model traces and processes them using a Graph Neural Network to produce a structural hallucination score. This signal is then integrated with entropy and confidence-based measures to form a unified reward signal for reasoning quality.

By combining structural reasoning analysis with uncertainty-aware feedback, our framework provides a more comprehensive mechanism for identifying unreliable reasoning and guiding LLM behavior.

2 Related work

Hallucination detection and mitigation in Large Language Models (LLMs) has become an important research area due to the tendency of models to generate confident yet incorrect information. Recent work has explored different approaches such as analyzing internal model signals, leveraging reasoning stability, and applying graph-based representations of computational traces.

3 Related work

3.1 Neural Message-Passing on Attention Graphs for Hallucination Detection

Frasca et al. propose a framework called CHARM that models hallucination detection as a graph learning problem. The authors represent the internal computation of large language models as attributed graphs, where tokens are treated as nodes and attention relationships form the edges between them. Node and edge attributes include attention scores and activation features extracted from the model. By applying Graph Neural Networks (GNNs) over these attention graphs, CHARM

captures complex interactions between tokens and model signals and achieves improved hallucination detection performance across multiple benchmarks.

Limitation. The method focuses primarily on hallucination detection and does not actively mitigate hallucinated outputs during generation.

3.2 Thinking, Faithful and Stable: Mitigating Hallucinations in LLMs

Zou et al. introduce a self-correcting framework designed to detect and mitigate hallucinations during multi-step reasoning processes in large language models. Instead of evaluating only the final answer, their method monitors intermediate reasoning signals to identify unreliable outputs. The framework utilizes two key uncertainty indicators: self-assessed confidence alignment and token-level entropy spikes. These signals are integrated into a reinforcement learning-based training procedure through a composite reward function that penalizes unjustified confidence and unstable reasoning behavior.

Limitation. While the approach improves reasoning stability, it does not analyze structural relationships within the model’s internal attention mechanisms.

3.3 SelfCheckGPT: Zero-Resource Hallucination Detection

Manakul et al. propose SelfCheckGPT, a sampling-based hallucination detection method that operates without access to internal model states. The approach generates multiple responses for a given prompt and measures their semantic consistency. If the responses diverge significantly, the generated information is likely hallucinated.

Limitation. The approach requires multiple generations for each prompt, increasing computational cost, and it relies solely on output consistency without leveraging internal model signals.

3.4 Chain-of-Verification for Hallucination Reduction

Dhuliawala et al. introduce the Chain-of-Verification (CoVe) framework, which reduces hallucinations through a structured verification process. The model first generates an answer, then produces verification questions, answers them independently, and revises the original response based on the verification results. This multi-step reasoning process improves factual reliability across several tasks.

Limitation. The verification pipeline requires multiple inference stages, increasing response latency, and relies on the same model for verification, which may propagate errors.

3.5 SelfCheckAgent for Hallucination Detection

Muhammed et al. propose SelfCheckAgent, a multi-agent framework for hallucination detection. The system employs multiple evaluation agents that analyze generated responses from different perspectives, including contextual reasoning and symbolic consistency, to identify hallucinated information.

Limitation. The framework introduces additional system complexity and does not analyze internal transformer representations such as attention structures.

3.6 Zero-Resource Hallucination Prevention

Luo et al. propose a hallucination prevention approach based on a self-evaluation mechanism that estimates the model’s familiarity with a topic before generating responses. If the model identifies insufficient familiarity with the prompt content, it avoids generating potentially hallucinated outputs.

Limitation. The method focuses on preventing generation rather than correcting hallucinated responses and does not incorporate structural reasoning signals from internal model computations.

4 Datasets

To be written during the Final submission.

5 Methodology

Our approach combines structural analysis of internal model computations with uncertainty-based reasoning signals to detect and mitigate hallucinations in Large Language Models (LLMs). The proposed framework consists of two components: (1) a hybrid pipeline that extracts structural representations from model traces and evaluates them using a Graph Neural Network (GNN), and (2) a unified reward function that integrates structural and uncertainty-based signals.

5.1 Hybrid Trace-Based Graph Pipeline

As the LLM generates reasoning steps, we extract multiple computational traces including token probabilities, attention matrices, and model activations. These signals are used to construct a dynamic attention graph representing the internal reasoning structure of the model.

Each generated token is treated as a node in the graph, while attention relationships between tokens define directed edges. Edge attributes correspond to attention weights, and node features include hidden representations or activation signals from the model. To reduce noise and computational complexity, the attention matrices are sparsified using a threshold τ , retaining only the most significant attention flows.

The resulting directed graph captures the structural dependencies among tokens during reasoning. A Graph Neural Network processes this graph to propagate information along attention edges and compute a structural hallucination score for the generated reasoning trajectory.

The GNN updates node representations through message passing across neighboring nodes according to:

$$h_i^{(t+1)} = \text{up}_t \left(h_i^{(t)}, \frac{1}{\deg_{in}(i)} \sum_{j:(i,j) \in E} \text{msg}_t \left(h_j^{(t)}, x_{E(i,j)}, p_{i,j} \right) \right)$$

where $h_i^{(t)}$ represents the hidden state of node i at layer t , E denotes the set of edges in the attention graph, and $\deg_{in}(i)$ is the in-degree of node i . The message and update functions learn structural patterns associated with hallucinated reasoning.

The final GNN output produces a continuous hallucination score $\hat{y}$ that reflects inconsistencies in the reasoning structure.

5.2 Unified Reward Function

To guide model behavior, we combine structural and uncertainty-based signals into a unified reinforcement learning reward. The final reward formulation integrates three complementary components: confidence calibration, entropy-based stability, and the structural hallucination score derived from the GNN.

The overall reward is defined as:

$$R_{\text{total}} = R_{\text{confidence}} + R_{\text{entropy}} - \gamma \cdot R_{\text{CHARM}}$$

where $R_{\text{confidence}}$ measures alignment between the model’s predicted confidence and actual correctness, R_{entropy} acts as a smoothing regularizer that discourages unstable token-level probability spikes, and R_{CHARM} represents the structural hallucination signal derived from the attention graph. The hyperparameter γ controls the influence of the structural critic.

This reward formulation provides triangulated feedback to the model: confidence calibration ensures reliable self-assessment, entropy regularization stabilizes token generation, and the graph-based critic evaluates structural reasoning coherence.

By integrating these complementary signals, the proposed framework encourages LLMs to generate reasoning trajectories that are both structurally consistent and uncertainty-aware.

6 Results

To be written during the Final submission.

References

Author Contributions

Authors of accepted papers are *encouraged* to include a statement that declares the individual contribution of every author, especially when there are co-authors that made equal contributions to the research.

A Appendix

Any possible appendices should be placed after bibliographies. If your paper has appendices, please submit the appendices together with the main body of the paper. There will be no separate supplementary material submission. The main text should be self-contained; reviewers are not obliged to look at the appendices when writing their review comments.